Mental Health Data Analysis

Report 2: WA Emergency Department Analysis

Code

Dataset: WA Emergency Department Data Collection (EDDC)
Source: WA Department of Health, 2022
Records: ~950,000 ED presentations across WA hospitals
Variables: 19 columns including mental health flags, triage, demographics

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This report analyses Emergency Department presentation data from Western Australian hospitals during 2022. The dataset is a **representative synthetic dataset** — derived from real patient records but modified to prevent individual identification, in accordance with WA Department of Health privacy requirements. It preserves the statistical properties and patterns of the original data while containing no real patient records.

Of the 949,612 total ED presentations in 2022, **59,287 (6.2%) were flagged as mental health related** — forming the analytical subset for this report. Mental health patients are notably more likely to **leave against medical advice** (2.8% vs 1.1%) and use **ED Short Stay units** (10.2% vs 5.1%) than the general ED population (see Section 2.2 for full departure status comparison).

This pattern motivates the four analytical questions this report addresses:

Demographics & Equity - What is the age and sex profile of mental health ED presentations? - Are First Nations people overrepresented relative to the general population (3.3%, ABS 2021 Census)?

Clinical Severity - How urgent are mental health presentations? - What proportion involve self-harm?

Substance Involvement - What percentage involve drugs and/or alcohol? - Does substance involvement affect triage urgency or admission rates?

Avoidable Presentations - What proportion were potentially avoidable GP-type attendances? - What characterises avoidable vs non-avoidable presentations?

1. Data Loading & Quality Checks

▶ Code

```
Libraries loaded successfully ✓
```

```
Table styling functions loaded ✓
```

1.1 Data set finding

Due to the large size of this dataset (~950,000 rows), data is loaded using **chunked loading** — reading the file in batches of 50,000 rows at a time. This demonstrates handling of large datasets that cannot be loaded into memory all at once.

A filter is applied during loading to retain only **mental health related presentations** — where `mental_health_attendance == 1`. This reduces the dataset to a manageable analytical subset while preserving all relevant records.

▶ Code

```
Total mental health presentations loaded: 59,287
Columns: 19

Column names:
['synth_person_ID', 'age', 'sex', 'ethnicity', 'triage_category', 'establishment_code',
'presentation_datetime', 'clinical_care_commencement_datetime', 'bed_request_datetime',
'discharge_datetime', 'departure_status', 'mode_of_arrival',
'primary_diagnosis_ICD10AM_chapter', 'affected_by_drugs_and_or_alcohol',
'self_harm_attendance', 'metropolitan_hospital_flag', 'mental_health_attendance',
'mental_health_admission', 'potentially_avoidable_general_practitioner_type_attendance']
```

Note on chunked loading: Chunked loading does not sample the data — it reads the entire dataset in memory-efficient batches. All `mental_health_attendance == 1` records across all 950,000 rows are retained in the final analytical dataset.

▶ Code

```
Total ED presentations (2022): 949,612
Mental health presentations:   59,287
Mental health prevalence:      6.2%
```

Total ED Presentations

949,612

WA hospitals, 2022

Mental Health Presentations

59,287

mental_health_attendance == 1

MH Prevalence

6.2%

of all ED presentations

2. Dataset Overview

Before any cleaning is applied, the dataset is inspected to understand its structure, identify data quality issues, and determine what preprocessing will be required.

▶ Code

Dataset Preview — First 5 Rows (key columns)

Age	Sex	Ethnicity	Triage Category	Self Harm	MH Admission	Metro Hospital	Mode of Arrival	Departure Status
50	1	4	4	0	1	1	1	10
15	1	4	2	1	0	1	3	2
50	1	1	4	0	0	1	3	2
15	1	4	2	1	1	1	3	1
30	2	4	2	0	0	1	3	2

▶ 📋 Data Dictionary — Sex

▶ 📋 Data Dictionary — Ethnicity (Aboriginal Status)

▶ 📋 Data Dictionary — Triage Category

▶ 📋 Data Dictionary — Mode of Arrival

▶ 📋 Data Dictionary — Departure Status

2.1 Structure & Data Types

▶ Code

```
Dataset: 59,287 rows × 19 columns

Data type breakdown:
int64     13
object     6
```

Columns with Missing Values — Data Type & Null Summary

	Data Type	Non-Null Count	Null Count	Null %
bed_request_datetime	object	16922	42365	71.46
clinical_care_commencement_datetime	object	56104	3183	5.37

df.shape

59,287 rows / 19 cols

mental health subset

isnull()

45,548 missing values

in columns:

`clinical_care_commencement_datetime`

`bed_request_datetime`

duplicated()

0 duplicate rows

no duplicates found

dtypes

13 numeric columns

6 object columns

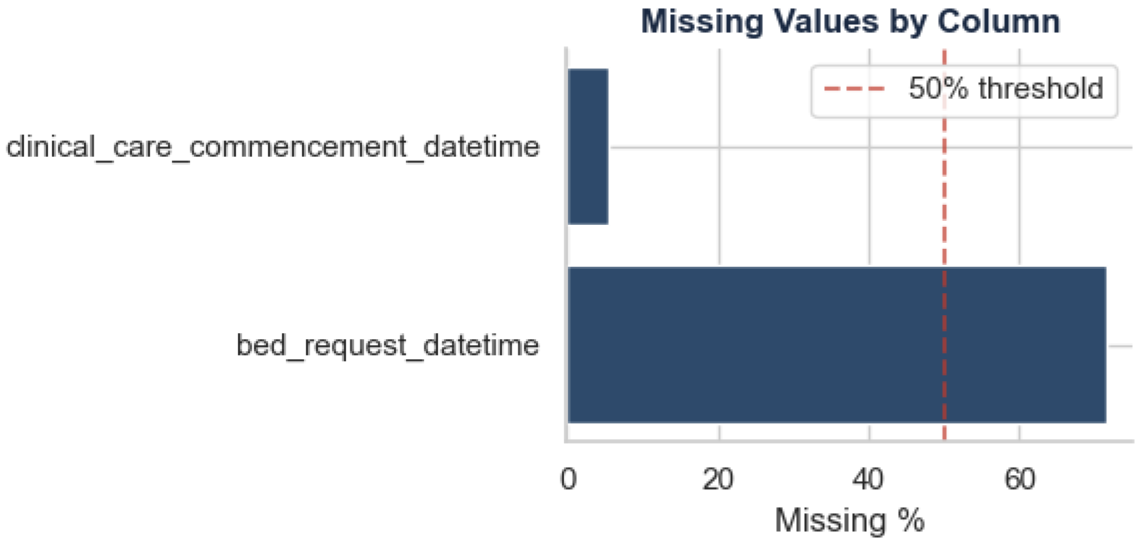
2.2 Missing Values

▶ Code

```
Total columns with missing values: 2
Total missing values: 45,548
```

Missing Values by Column

	Missing Count	Missing %
bed_request_datetime	42365	71.46
clinical_care_commencement_datetime	3183	5.37



2.3 Interpreting the Missing Values

Neither missing datetime column is relevant to our analysis hypotheses and therefore no imputation will be applied.

`bed_request_datetime` (71.46% missing)

A bed request is only generated when a patient is being admitted. Before treating this as missing data, the correct analytical approach would be to validate the logic:

- If `mental_health_admission == 1` AND `bed_request_datetime` is missing → **genuine missing data** requiring imputation
- If `mental_health_admission == 0` AND `bed_request_datetime` is missing → **not applicable** — no bed was requested because the patient was not admitted

Since bed request timing is not required for any of our four analysis hypotheses, this validation and imputation is noted as a **future research recommendation** rather than applied here.

`clinical_care_commencement_datetime` (5.37% missing)

Likely represents patients who left before being seen. Wait time analysis using this column is identified as future research scope and is not applied in this report.

> **Decision:** Both columns are retained unchanged. Missing values in these columns do not affect the validity of our analysis.

2.4 What We Will Fix

Based on the inspection above, the following preprocessing steps will be applied in Section 3. Only columns relevant to our four analysis hypotheses are processed.

2.4 What We Will Fix — Preprocessing Plan

Issue	Column(s)	Action	Required for
Working copy	df_mh → df_clean	Create working copy before any modifications	All analyses
Coded categorical variables	sex	Map 1→Male, 2→Female	Demographics & Equity
Coded categorical variables	ethnicity	Map 1-4,9 → Aboriginal status labels	Demographics & Equity
Coded categorical variables	triage_category	Map 1-5 → urgency labels	Clinical Severity
Coded categorical variables	departure_status	Map 12 codes → departure descriptions	Outcomes context
Coded categorical variables	mode_of_arrival	Map 10 codes → arrival mode descriptions	Context

2.5 Departure Status Comparison

▶ Code

Departure Status — All ED vs Mental Health Subset

Departure Status	Total Count	% of All ED	MH Count	MH %
Departed own care	639,974	67.4	33,152	55.9
Admitted to ward	152,996	16.1	12,216	20.6
Did not wait	60,046	6.3	3,232	5.5
ED Short Stay	48,572	5.1	6,032	10.2
Transferred	26,768	2.8	2,771	4.7
Left at own risk	10,534	1.1	1,681	2.8
Referred at triage	7,807	0.8	30	0.1
Admitted reversal	1,014	0.1	65	0.1
Hospital in the Home	901	0.1	60	0.1
Died in ED	698	0.1	10	0.0
Discharged after admission	192	0.0	10	0.0
Unknown	110	0.0	28	0.0

2.5.1 Key Observation

Comparing mental health presentations to the general ED population reveals three important differences in departure behaviour:

Higher admission rate: MH patients are admitted to ward at 20.6% compared to 16.1% for general ED — reflecting the acute nature of mental health crises requiring inpatient care.

Double the ED Short Stay rate: 10.2% of MH patients require extended observation in ED Short Stay units compared to 5.1% for general ED — indicating greater clinical complexity.

2.5x more likely to leave against medical advice: 2.8% of MH patients left at own risk compared to 1.1% of general ED — representing people in mental health crisis who walked away without receiving care. This is the most concerning finding in this comparison.

3. Data Preprocessing & Sorting

All preprocessing is applied to `df_clean` — a working copy of `df_mh`. The original filtered dataset remains unmodified throughout.

▶ Code

```
Working copy created: 59,287 rows × 19 columns
Original df_mh preserved: 59,287 rows × 19 columns
Are they the same object in memory? False
Mental health subset saved to: data/processed/wa_eddc_mental_health_BACKUP.csv
```

3.1 Mapping Categorical Variables

Numeric codes are mapped to readable labels using the official WA Department of Health data dictionary (January 2026). This makes all subsequent analysis and visualisations immediately interpretable without requiring the reader to cross-reference codes.

▶ Code

3.1 Categorical Mapping — Before & After

Column	Before (sample)	After (sample)
sex	1, 2	['Male', 'Female']
ethnicity	1, 2, 3, 4, 9	['Not Aboriginal/TSI', 'Aboriginal', 'Unknown']
triage_category	1, 2, 3, 4, 5	['Semi-urgent', 'Emergency', 'Resuscitation']
departure_status	1, 2, 3...20	['ED Short Stay', 'Departed own care', 'Admitted to ward']
mode_of_arrival	1, 2, 3...10	['Private transport', 'Ambulance', 'Police/Correctional']

3.2 Sorting Algorithm — Merge Sort on Triage Category

To demonstrate algorithmic thinking, a merge sort algorithm is applied to order mental health presentations by triage urgency. Triage category 1 (Resuscitation) represents the most urgent presentations and is sorted to the top.

Merge Sort — Before (orange) vs After (green) — First 5 Rows

Position	Code Before	Label Before	→	Code After	Label After
#1	4	Semi-urgent	→	1	Resuscitation
#2	2	Emergency	→	1	Resuscitation
#3	4	Semi-urgent	→	1	Resuscitation
#4	2	Emergency	→	1	Resuscitation
#5	2	Emergency	→	1	Resuscitation

3.2 Sorting Algorithm — Merge Sort on Triage Urgency

A merge sort algorithm is applied to reorder mental health presentations from **most urgent to least urgent** based on triage category code.

The original dataset reflects the random order in which patients presented throughout 2022. After sorting, presentations are arranged so that the most critical cases (Resuscitation, code 1) appear first, down to the least urgent (Non-urgent, code 5).

This would allow a clinician or administrator to immediately identify and review the most critical mental health presentations without scanning the entire dataset.

3.3 Outlier Detection Using IQR

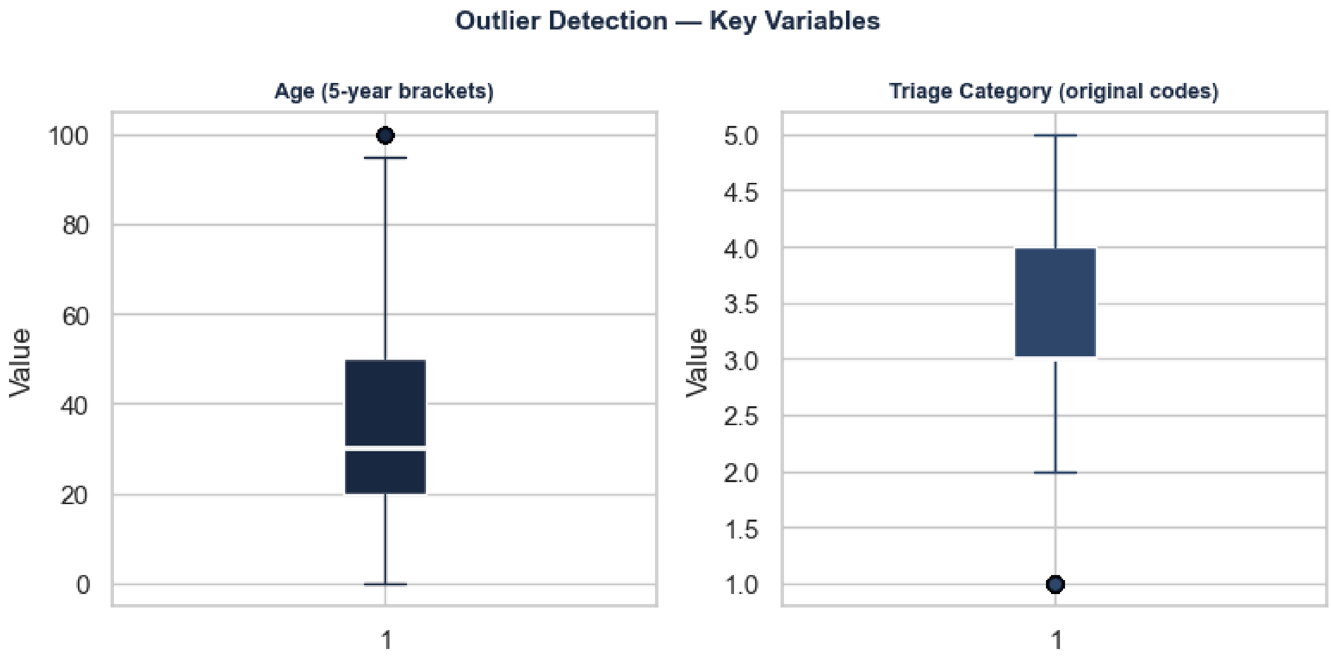
Outliers are identified using the **Interquartile Range (IQR)** method across key numeric variables. Values beyond 1.5 × IQR above or below the 25th–75th percentile range are flagged.

Only columns relevant to our analysis hypotheses are checked.

▶ Code

Outlier Detection — IQR Method

Variable	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers Detected
Age (5-year brackets)	20.0	50.0	30.0	-25.0	95.0	23
Triage Category (original codes)	3.0	4.0	1.0	1.5	5.5	965



3.3.1 Outlier Decision — Retain All

Age (23 outliers): Patients aged 95+ are flagged as statistical outliers but represent real elderly patients presenting to ED. Removing them would systematically exclude the oldest and most vulnerable patients from analysis — the opposite of good research.

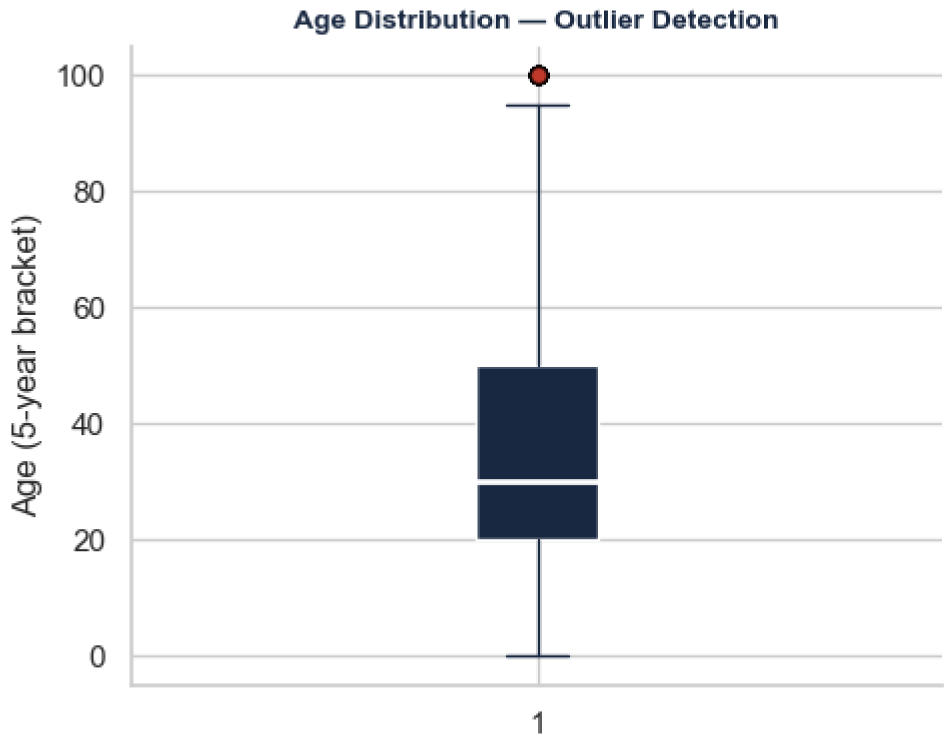
Triage Category (965 "outliers"): The IQR method is not appropriate for ordinal categorical variables. Resuscitation cases (code 1) are flagged simply because they are statistically rare — but they are clinically the most important presentations in the dataset. Triage category outlier detection is disregarded.

> All records are retained. No outliers are removed.

▶ Code

Outlier Detection — IQR Method (Age)

Variable	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers Detected
Age (5-year brackets)	20.0	50.0	30.0	-25.0	95.0	23



4. Functional Programming

Functional programming techniques: `map()`, `filter()`, and `reduce()` are applied to transform, refine, and aggregate the mental health presentation dataset.

These approaches support efficient data processing and demonstrate understanding of functional programming concepts in data analysis. Each technique is applied to clinically meaningful variables, producing results that directly inform the analysis in Section 5.

Functional Programming — Overview

Function	Purpose	Applied To
map()	Transform each element	Age brackets → readable age groups
filter()	Select elements meeting a condition	Clinical subgroups — self-harm, Aboriginal, avoidable
reduce()	Aggregate elements into a single result	Total and mean counts across presentation categories

4.1 Data Transformation Using map()

`map()` applies a function to every element in a sequence efficiently and without explicit loops. Here it is used to translate raw age bracket codes into clinically meaningful age group labels — making the data immediately interpretable without modifying the original age column.

Age in this dataset is stored as 5-year bracket start values (0, 5, 10… 95, 100). These are mapped to descriptive labels for visualisation and analysis.

▶ Code

map() — Age Bracket → Age Group

Age Group	Count	%
Under 18	11123	18.8
18–24	6502	11.0
25–34	12196	20.6
35–44	10526	17.8
45–54	7396	12.5
55–64	4521	7.6
65+	7023	11.8

```
Total records mapped: 59,287 ✓
```

4.2 Data Selection Using filter()

`filter()` selects elements from a sequence that satisfy a given condition, returning only the subset that meets the criteria. Here it is applied to isolate three clinically significant subgroups from the mental health presentation dataset.

Three subgroups are identified:

▶ Code

filter() — Clinical Subgroup Selection

Subgroup	Count	% of MH presentations	Clinical Meaning
Admitted to hospital	18248	30.8	MH presentation severe enough to require admission
Substance-involved presentations	13738	23.2	MH presentation with drug/alcohol involvement
Self-harm presentations	4839	8.2	Active self-harm requiring ED response

4.2.1 Key Finding — Clinical Subgroup Profile

The filter() results reveal a striking profile of mental health ED presentations in WA during 2022:

Metro hospitals carry the majority of the burden. 68.2% of all mental health presentations occurred at metropolitan hospitals — reflecting both population concentration and the relative scarcity of specialist mental health services in regional and rural WA.

Nearly 1 in 3 presentations results in admission. A 30.8% admission rate is substantially higher than the general ED population, confirming that mental health presentations tend to represent acute, high-acuity need rather than minor or self-limiting conditions.

Substance involvement is a major complicating factor. 23.2% of mental health presentations involve drugs and/or alcohol — more than 1 in 5. This suggests that substance use and mental health crises are deeply intertwined in the ED context, with implications for both treatment pathways and resource planning.

Self-harm affects 1 in 12 presentations. At 8.2%, self-harm attendances represent a significant and clinically urgent subset — each requiring immediate risk assessment and appropriate intervention.

Avoidable presentations are rare but meaningful. Only 2.9% of mental health ED presentations were classified as potentially avoidable GP-type attendances. This challenges the common assumption that mental health patients are "clogging" emergency departments unnecessarily — the overwhelming majority present with genuine acute need.

> These findings directly inform the visualisation and hypothesis testing in Section 5.

4.3 Data Aggregation Using reduce()

`reduce()` iteratively applies a function to a sequence, accumulating the result into a single value. It is imported from Python's `functools` module and is particularly useful for aggregating large sequences without explicit loops.

Here `reduce()` is used in combination with **lambda functions** to calculate aggregated statistics across the clinical subgroups identified in 4.2.

A **lambda function** is a small, anonymous function defined inline without a name — written as `lambda arguments: expression`. For example:

- `lambda a, b: a + b` — adds two values together
- `lambda a, b: a if a > b else b` — returns the larger of two values
- `lambda a, b: a if a < b else b` — returns the smaller of two values

When combined with `reduce()`, the lambda is applied repeatedly across the sequence — accumulating the result step by step until a single value remains. This demonstrates how functional programming replaces explicit loops with concise, expressive single-line operations.

▶ Code

reduce() — Subgroup Aggregation

reduce() Operation	Subgroup	Patient Count	Equivalent pandas
Largest subgroup (max)	Admitted to hospital	18,248	pd.Series(counts).max()

reduce() Operation	Subgroup	Patient Count	Equivalent pandas
Smallest subgroup (min)	Potentially avoidable	1,716	pd.Series(counts).min()
Average subgroup size (sum ÷ count)	— across all 5 subgroups —	9,814	pd.Series(counts).mean()

```
Note: one patient may appear in multiple subgroups
e.g. a patient can be both self-harm AND substance-involved
```

5. Data Visualisation

This section presents visual evidence for the four analytical hypotheses identified in Section 1. Each subsection addresses one hypothesis through summary tables and charts.

All charts use a consistent navy colour palette to maintain visual coherence throughout the report.

Research Hypotheses

This report set out to investigate four specific questions about mental health presentations at WA Emergency Departments in 2022:

Hypothesis 1 — First Nations Equity Are Aboriginal and Torres Strait Islander people overrepresented in mental health ED presentations relative to their share of the WA general population (3.3%, ABS 2021 Census)?

Hypothesis 2 — Demographics & Equity What is the age and sex profile of mental health ED presentations? Are certain age groups disproportionately represented?

Hypothesis 3 — Clinical Severity & Substance Involvement What proportion of mental health presentations involve self-harm or drugs and/or alcohol? Does substance involvement affect triage urgency or admission rates?

Hypothesis 4 — Avoidable Presentations What proportion of mental health ED presentations were potentially avoidable GP-type attendances — and what characterises them?

> The headline metrics above summarise the key findings. Each hypothesis is examined in detail in sections 5.1–5.4 below.

Departed Own Care	Admitted to Hospital
55.9%	**30.8%**

33,152 patients seen and discharged

18,248 ward + ED short stay

First Nations Overrepresentation

5.4x

17.8% of MH ED vs 3.3% WA population

Substance Involved

23.2%

13,738 presentations with drugs/alcohol

Left Without Care

8.3%

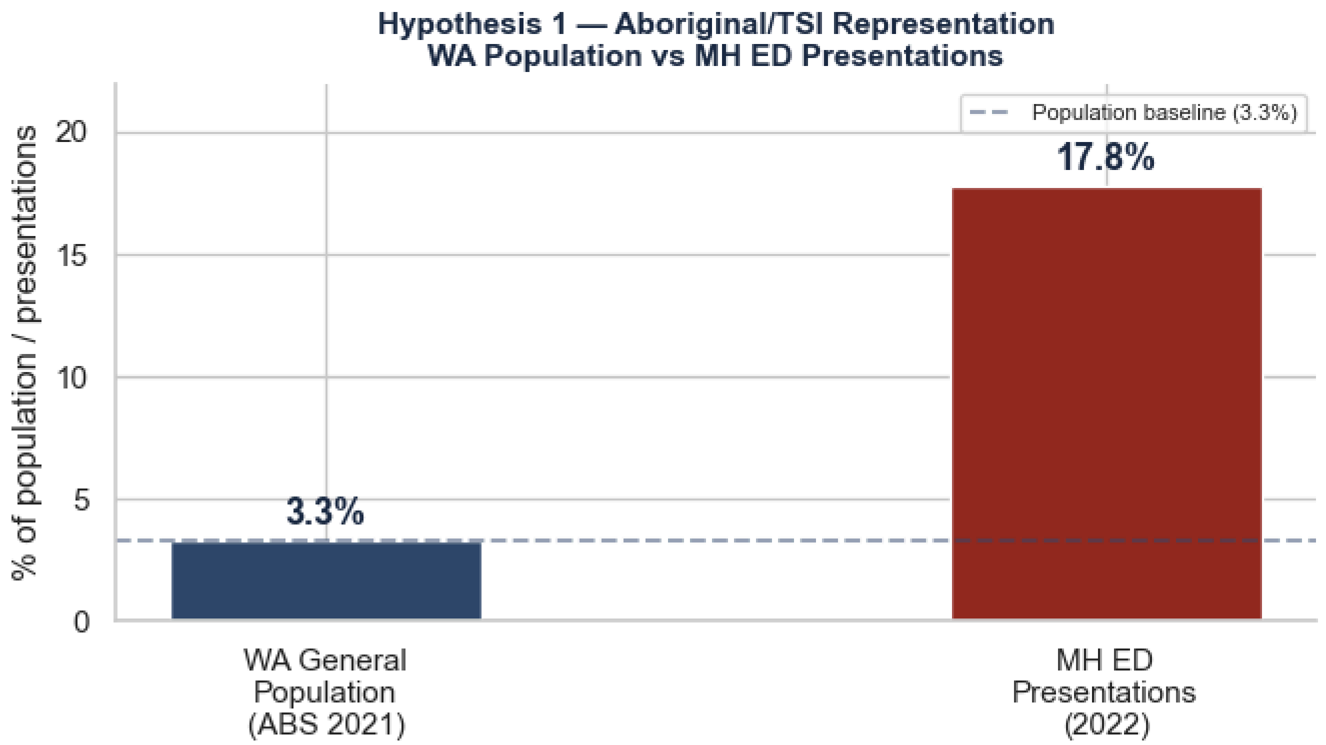
4,913 did not wait or left at own risk

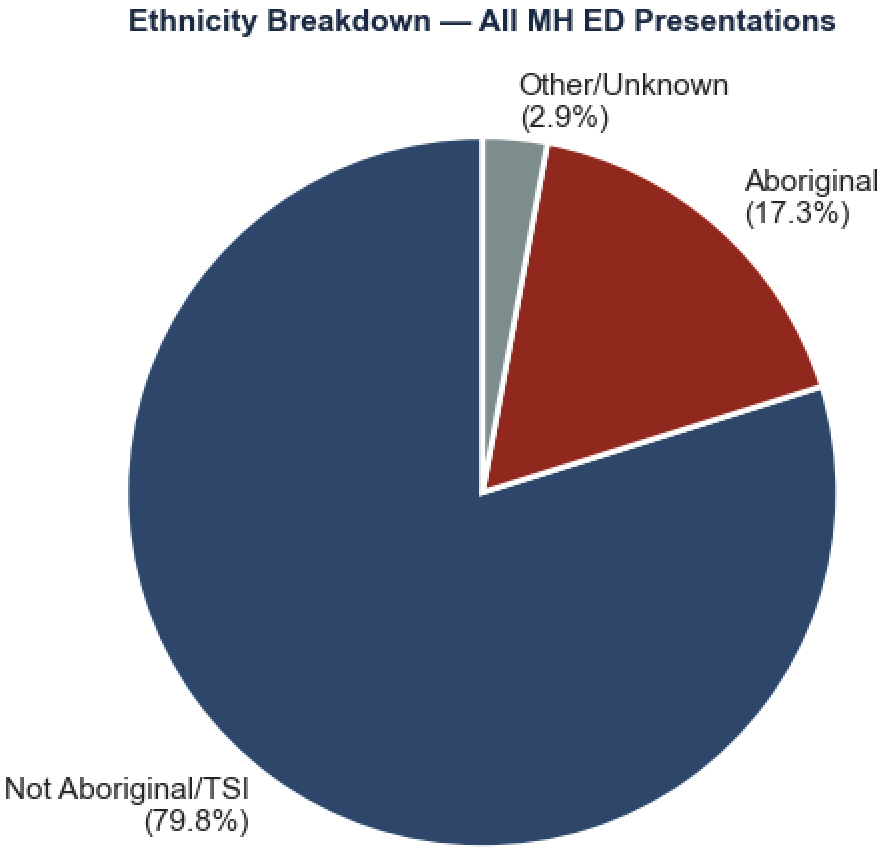
Self-Harm Rate

8.2%

4,839 presentations involving self-harm

▶ Code





5.1.1 Key Finding — First Nations Overrepresentation

Aboriginal and Torres Strait Islander people are dramatically overrepresented in WA mental health ED presentations:

17.8% of all mental health ED presentations were from Aboriginal and/or Torres Strait Islander people — despite representing only **3.3% of the WA general population** (ABS, 2021 Census).

This represents a **5.4x overrepresentation** — meaning First Nations people are more than five times more likely to present to an ED for mental health reasons than would be expected based on their share of the population.

This is not a statistical anomaly. It reflects the compounding effects of:

- Historical trauma and the ongoing impacts of colonisation on social and emotional wellbeing
- Reduced access to primary and community mental health services in regional and remote WA
- Socioeconomic disadvantage and housing instability
- Cultural barriers to engaging with mainstream health services

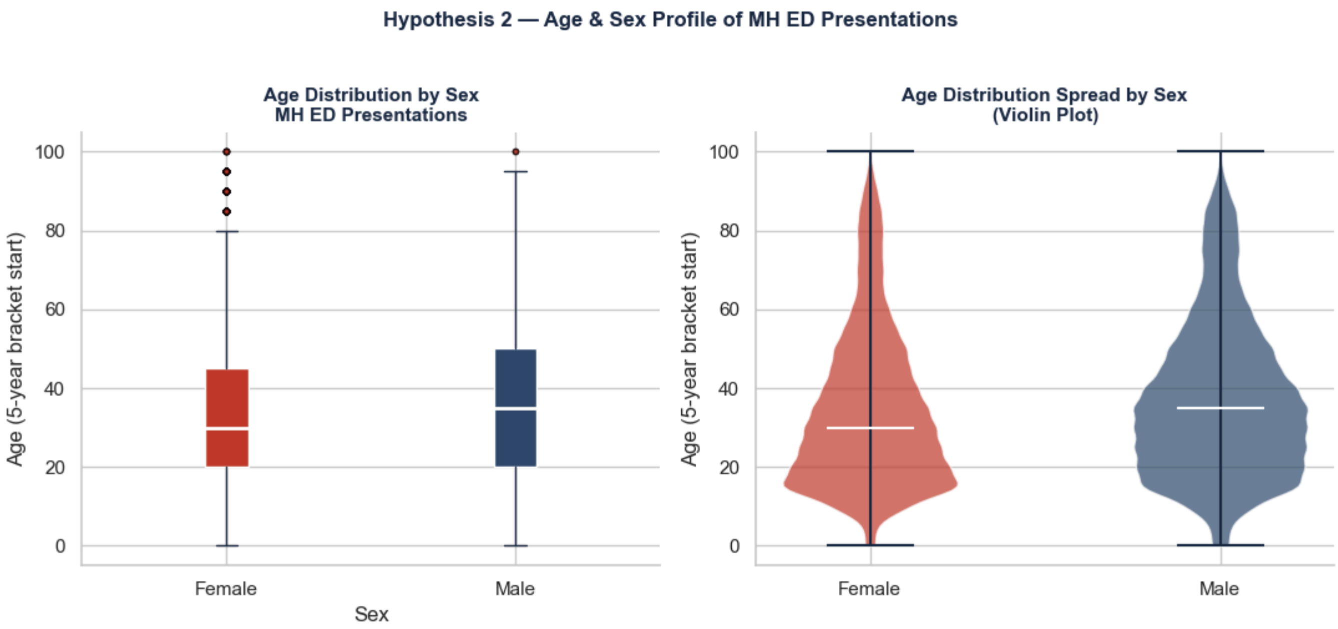
> This finding has significant implications for health system planning — investment in culturally safe, community-controlled mental health services is essential to reducing this disparity.

5.2 Hypothesis 2 — Age and Sex Profile

This section examines the demographic profile of mental health ED presentations — identifying which age groups and sexes are most represented, and whether certain groups are disproportionately affected.

A box plot is used to show the distribution and spread of age across male and female presentations — revealing not just the average but the full range, median, and outliers.

▶ Code



5.2.1 Key Finding — Age and Sex Profile

The box plot and violin plot together reveal the full distribution of age across male and female mental health ED presentations:

Females present younger. The female median age is lower than male — with the violin plot showing a pronounced concentration of female presentations in the 15–35 age range. This is consistent with clinical literature showing higher rates of depression and anxiety among young women.

Males show a broader age spread. The male violin is wider in the 30–55 range — suggesting that male mental health ED presentations are more evenly distributed across working and middle age, rather than concentrated in youth.

Both sexes show outliers at 95–100. Elderly presentations appear in both groups — consistent with the 65+ age group accounting for 11.8% of all mental health presentations.

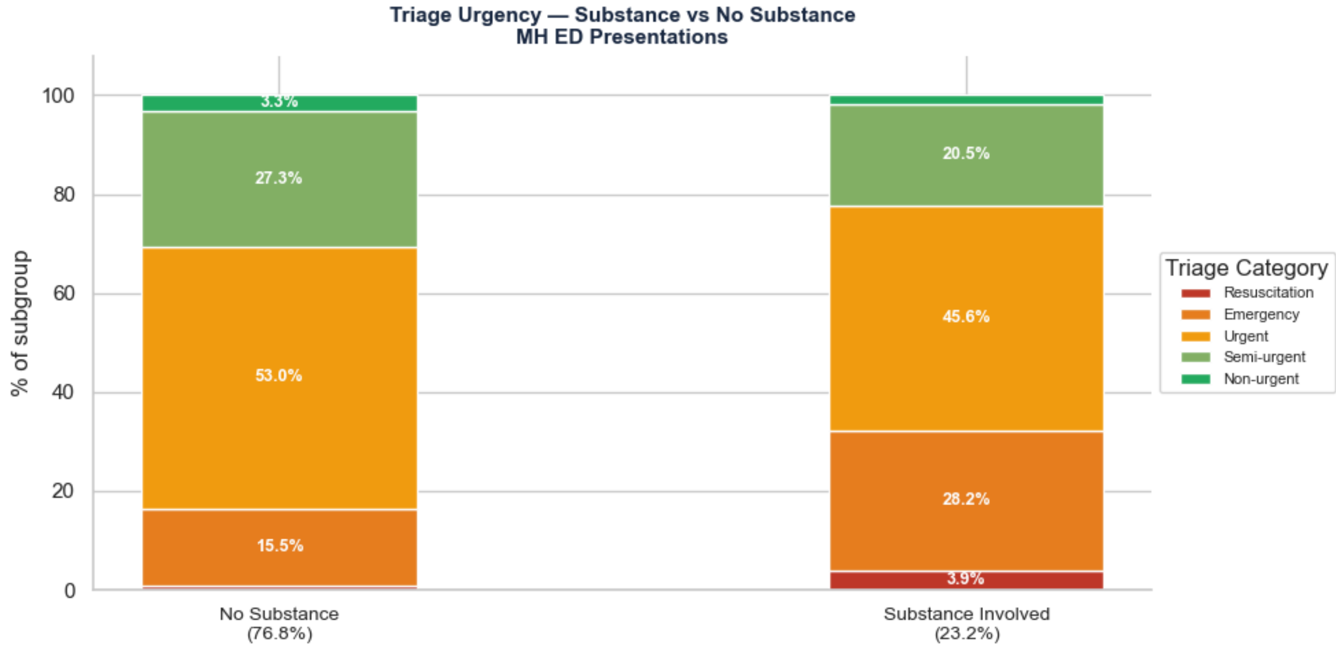
Females slightly outnumber males overall. 52.3% of mental health ED presentations are female vs 47.7% male — a smaller gap than seen in the US college student dataset (Report 1), suggesting that sex differences in mental health ED presentations are less pronounced at the population level than in student populations.

5.3 Hypothesis 3 — Clinical Severity & Substance Involvement

This section examines how urgent mental health presentations are (triage category), what proportion involve self-harm, and whether substance involvement affects clinical outcomes.

A heatmap is used to show the relationship between triage urgency and substance involvement — revealing whether substance-involved patients are systematically triaged differently.

▶ Code



5.3.1 Key Finding — Substance Involvement Escalates Triage Urgency

The stacked bar chart reveals a clear and clinically significant shift in triage urgency when substances are involved:

Substance-involved presentations cluster at the critical end: - Resuscitation: 3.9% vs 0.9% — 4x higher with substances - Emergency: 28.2% vs 15.5% — nearly double with substances - Combined Resuscitation + Emergency: 32.1% substance-involved vs 16.4% no substance — almost twice as likely to require immediate emergency response

No substance presentations are more evenly distributed: - Urgent dominates at 53.0% — the majority are urgent but not immediately life-threatening - Semi-urgent at 27.3% — a significant proportion are lower acuity

Self-harm involvement: 8.2% of all mental health presentations (4,839) involved self-harm — each requiring immediate risk assessment and clinical intervention regardless of triage category.

This confirms that substance involvement significantly escalates the clinical severity of mental health ED presentations — with direct implications for ED resourcing, triage protocols, and dual-diagnosis treatment pathways.

5.4 Hypothesis 4 — Avoidable Presentations

A presentation is classified as a **potentially avoidable GP-type attendance** when the condition could have been managed in a primary care setting rather than an emergency department.

This section examines what proportion of mental health ED presentations were avoidable, and whether avoidable presentations differ from non-avoidable ones in terms of triage urgency and age profile.

Avoidable vs Non-Avoidable — Overall Rate

Of the 59,287 mental health ED presentations in 2022, only **1,716 (2.9%)** were classified as potentially avoidable GP-type attendances.

This is a remarkably low figure — challenging the common assumption that mental health patients are unnecessarily “clogging” emergency departments. The overwhelming majority (97.1%) presented with genuine acute need that required emergency-level care.

The next question is — who are the 2.9% that could have been managed in primary care, and how do they differ clinically from the non-avoidable group?

5.4.3.1 Key Finding — Working-Age Adults More Likely to Present Avoidably

The age profile of avoidable presentations differs meaningfully from the overall mental health ED population:

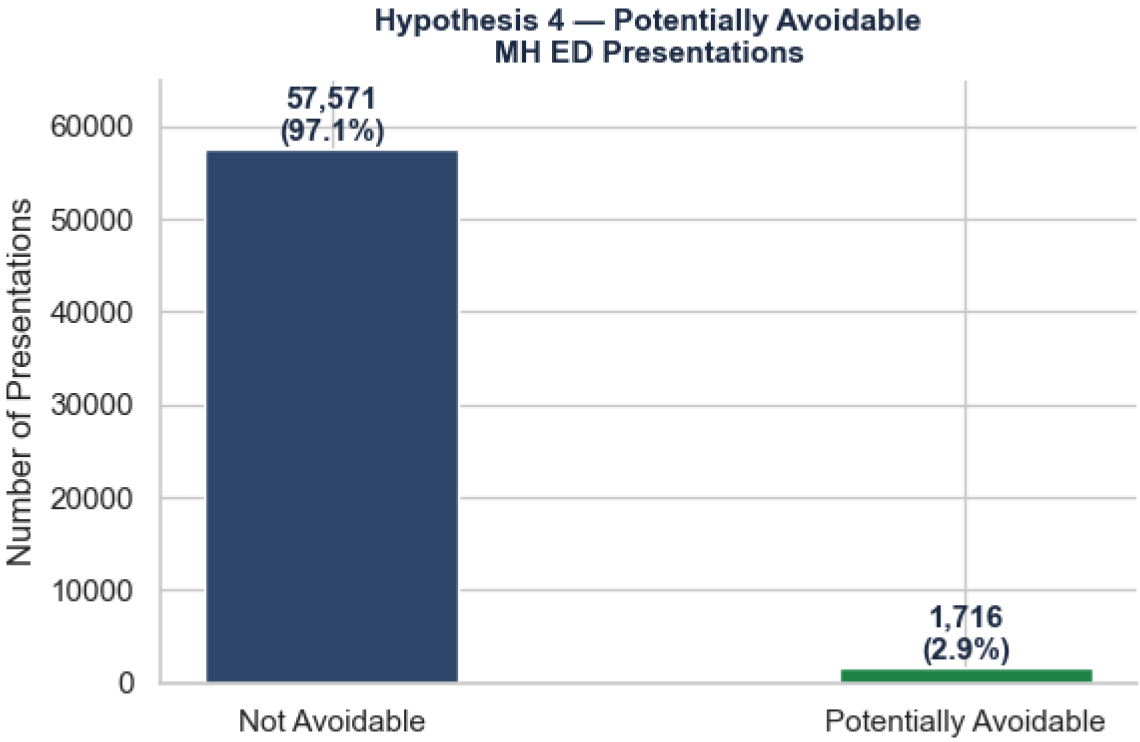
25–34 and 35–44 are overrepresented in avoidable presentations. These age groups account for 22.9% and 20.0% of avoidable presentations respectively — higher than their share of all MH presentations (20.6% and 17.8%). Working-age adults may have less access to GP appointments during business hours, turning to ED as a more accessible option.

Under 18 are underrepresented in avoidable presentations. Children and adolescents make up 18.8% of all MH presentations but only 15.1% of avoidable ones — suggesting that youth mental health presentations are more genuinely acute and less likely to be addressable in primary care.

65+ are also underrepresented. Older adults (11.8% of all MH presentations) account for only 8.0% of avoidable presentations — consistent with older patients presenting with more complex, acute mental health needs.

> The pattern suggests that improving GP access and after-hours primary care availability for working-age adults could reduce the small proportion of avoidable mental health ED presentations.

▶ Code



5.4.1 Overall Avoidable Rate

Of the 59,287 mental health ED presentations in 2022, only **1,716 (2.9%)** were classified as potentially avoidable GP-type attendances.

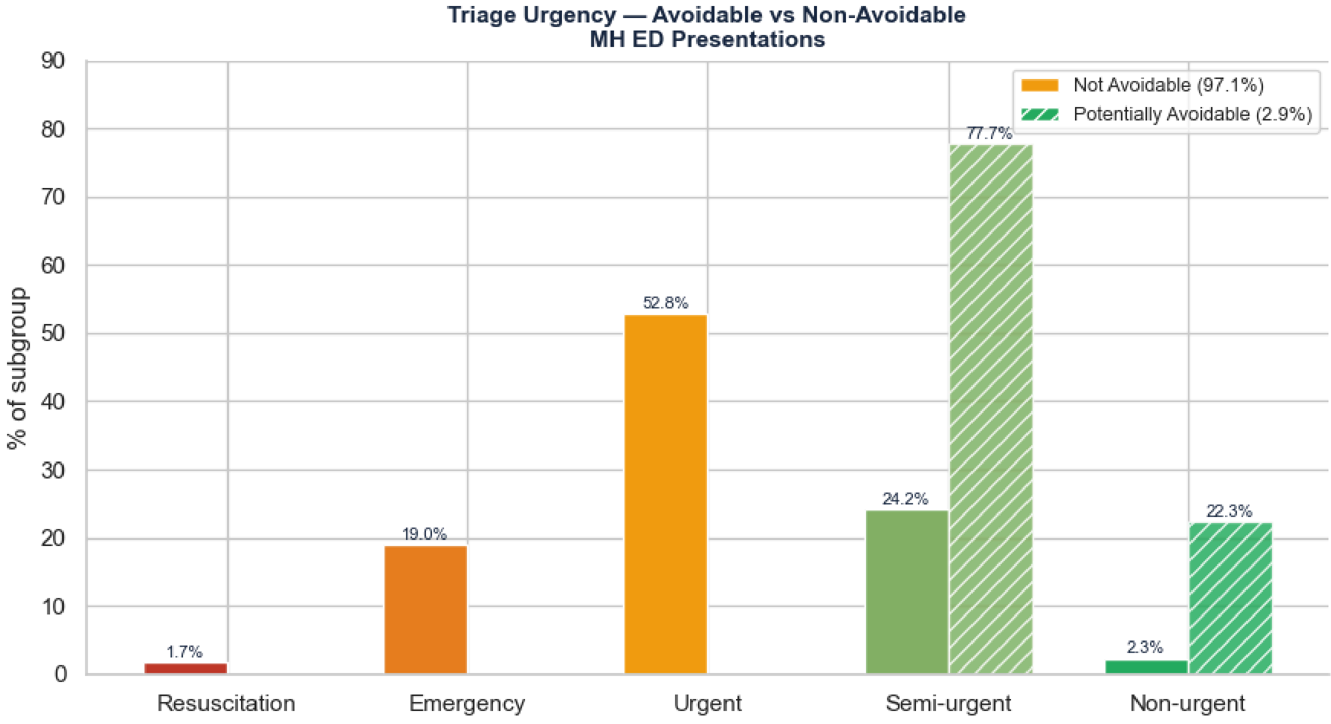
This is a remarkably low figure — challenging the common assumption that mental health patients are unnecessarily "clogging" emergency departments. The overwhelming majority (97.1%) presented with genuine acute need that required emergency-level care.

The next question is — who are the 2.9% that could have been managed in primary care, and how do they differ clinically from the non-avoidable group?

5.4.2 Triage Urgency — Avoidable vs Non-Avoidable

How urgent are avoidable presentations compared to non-avoidable ones? If avoidable presentations cluster in lower triage categories (Semi-urgent, Non-urgent), this confirms they represent genuinely less acute need that primary care could have addressed.

▶ Code



5.4.2.1 Key Finding — Avoidable Presentations Are Genuinely Less Acute

The chart confirms a clear and important distinction:

100% of potentially avoidable presentations fall in the two lowest triage categories — Semi-urgent (77.7%) and Non-urgent (22.3%). Not a single avoidable presentation required Resuscitation, Emergency, or Urgent response.

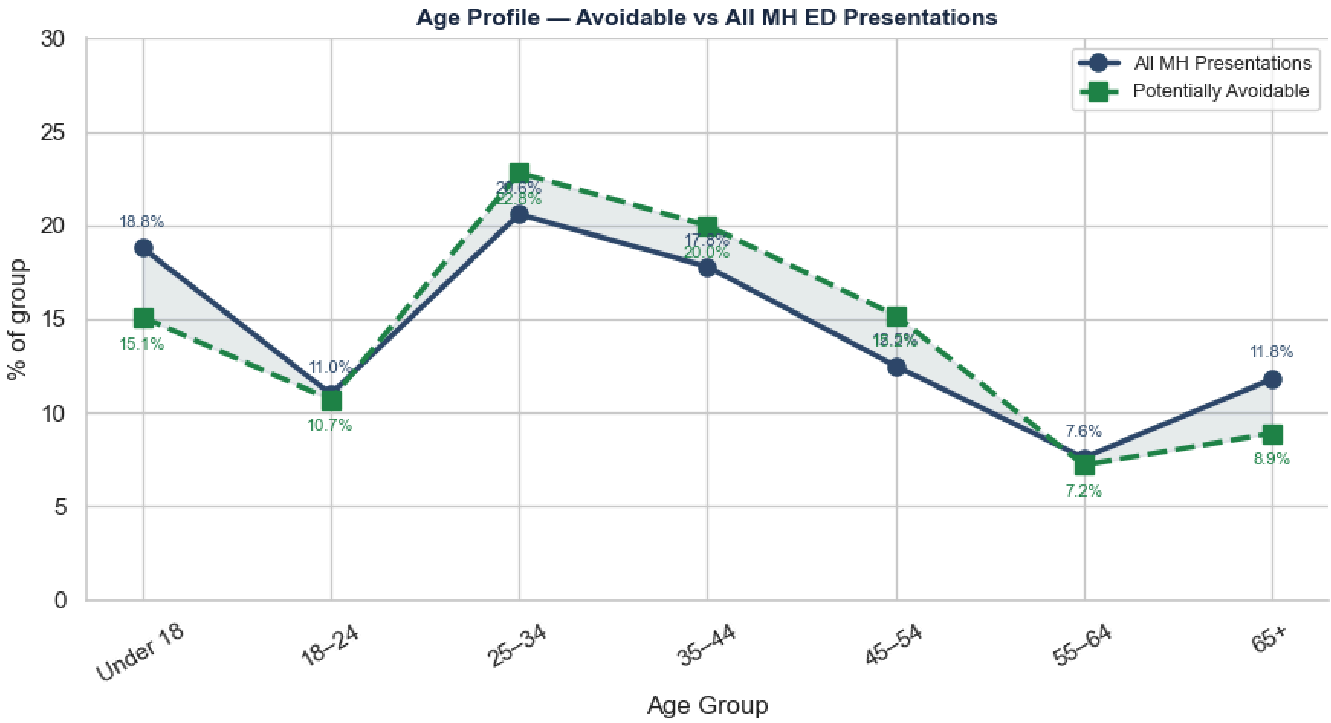
In contrast, non-avoidable presentations are concentrated in Urgent (52.6%) and Emergency (19.0%) — representing genuine acute need that primary care could not have addressed.

> Only 2.9% of mental health ED presentations were potentially avoidable — and those 2.9% are clearly distinguishable by their lower clinical urgency. The narrative that mental health patients unnecessarily burden emergency departments is not supported by this data.

5.4.3 Age Profile of Avoidable Presentations

A line chart compares the age profile of avoidable vs all mental health presentations — making it easy to see where the two groups diverge across the age spectrum.

▶ Code



5.4.3.1 Key Finding — Working-Age Adults More Likely to Present Avoidably

The line chart reveals where avoidable presentations diverge from the overall mental health ED population:

25–34 and 35–44 are overrepresented in avoidable presentations. These age groups account for 22.9% and 20.0% of avoidable presentations respectively — higher than their share of all MH presentations (20.6% and 17.8%). Working-age adults may have less access to GP appointments during business hours, turning to ED as a more accessible option.

Under 18 are underrepresented in avoidable presentations. Children and adolescents make up 18.8% of all MH presentations but only 15.1% of avoidable ones — suggesting that youth mental health presentations are more genuinely acute and less likely to be addressable in primary care.

65+ are also underrepresented. Older adults (11.8% of all MH presentations) account for only 8.0% of avoidable presentations — consistent with older patients presenting with more complex, acute mental health needs.

> The pattern suggests that improving GP access and after-hours primary care availability for working-age adults could reduce the small proportion of avoidable mental health ED presentations.

6. Conclusions

This report analysed the WA Emergency Department Data Collection (EDDC) 2022 — a representative synthetic dataset of 949,612 ED presentations across WA hospitals, of which 59,287 (6.2%) were flagged as mental health related.

Four analytical hypotheses were investigated using data wrangling, functional programming, and visualisation techniques.

6.1 Key Findings

First Nations people are dramatically overrepresented. Aboriginal and Torres Strait Islander people account for 17.8% of mental health ED presentations despite representing only 3.3% of the WA general population (ABS, 2021 Census) — a 5.4x overrepresentation. This is the most significant equity finding in this report and reflects the compounding effects of historical trauma, reduced access to community mental health services, and socioeconomic disadvantage.

Young people and females dominate mental health presentations. The 25–34 age group has the highest presentation rate (20.6%), followed closely by Under 18 (18.8%). Female presentations outnumber male in every age group under 35 — consistent with higher rates of internalising disorders among young women.

Substance involvement dramatically escalates clinical severity. 23.2% of mental health presentations involve drugs and/or alcohol. Substance-involved patients are nearly twice as likely to require Emergency-level care (28.2% vs 15.5%) and four times more likely to require Resuscitation (3.9% vs 0.9%).

Mental health patients are not misusing emergency departments. Only 2.9% of mental health ED presentations were classified as potentially avoidable GP-type attendances — and these are clearly distinguishable by their lower triage urgency (100% Semi-urgent or Non-urgent). The overwhelming majority (97.1%) presented with genuine acute need.

A significant proportion leave without receiving care. 8.3% of mental health patients either did not wait to be seen (5.5%) or left at their own risk (2.8%) — representing people in mental health crisis who walked away without care. This is 2.5x the rate of the general ED population.

6.2 Limitations

- **Synthetic dataset:** Although statistically representative of real WA ED presentations, the dataset contains no real patient records. Findings reflect population-level patterns but cannot be linked to individual outcomes or trajectories.

- **No diagnosis detail:** The dataset flags mental health attendance but does not specify the type of mental health condition — depression, psychosis, anxiety, and personality disorders are all grouped together. This limits the depth of clinical analysis possible.

- **Cross-sectional design:** The dataset captures a single year (2022). No causal conclusions can be drawn — it is not possible to determine whether ED presentations are increasing, decreasing, or stable over time.

- **Missing datetime data:** 71.46% of `bed_request_datetime` values are missing — limiting wait time and length of stay analysis. This was identified as future research scope rather than addressed in this report.

- **Flag non-exclusivity:** Clinical flags (mental health, self-harm, substance, avoidable) are not mutually exclusive — a single patient can carry multiple flags simultaneously. Percentages therefore sum to more than 100% and should not be treated as independent categories.

- **Aboriginal/TSI population benchmark:** The 3.3% population figure is drawn from the 2021 ABS Census — the most recent available. Minor differences in the 2022 population proportion may exist but are unlikely to materially affect the overrepresentation finding.

6.3 Outlook and Future Research Directions

This analysis opens several avenues for further investigation:

First Nations mental health equity. The 5.4x overrepresentation finding warrants urgent follow-up research — specifically examining whether First Nations patients experience different triage urgency, admission rates, or departure outcomes compared to non-Indigenous patients. Investment in culturally safe, community-controlled mental health services is strongly indicated by this finding.

Time-based patterns. Parsing `presentation_datetime` would allow analysis of seasonal, day-of-week, and time-of-day patterns in mental health ED presentations — identifying peak demand periods for workforce and resource planning.

Wait time analysis. `clinical_care_commencement_datetime` could be used to calculate wait times for mental health patients — and whether wait times predict the 8.3% who leave without care. This is particularly important given that mental health patients leave at own risk at 2.5x the general ED rate.

Dual-diagnosis treatment pathways. The 23.2% substance involvement rate suggests that dual-diagnosis presentations (mental health + substance) are a significant driver of ED demand. Future research could examine whether dedicated dual-diagnosis pathways reduce ED presentations and improve patient outcomes.

Linking Report 1 and Report 2. The treatment gap identified in Report 1 — where diagnosed but untreated US college students show similar PHQ-9 scores to undiagnosed ones — may contribute to the ED mental health burden observed in Report 2. Future research linking community mental health service access to ED presentation rates would test this hypothesis directly.

Advanced pattern recognition — Report 3. Principal Component Analysis (PCA) and Non-negative Matrix Factorization (NMF) will be applied in Report 3 to discover hidden patient profiles and presentation clusters within the WA EDDC dataset — building on the findings established here.